

Tarea S9.01. Consultas con MongoDB

Nivel 1

Crea una base de datos con MongoDB utilizando como colecciones los archivos adjuntos.

The screenshot shows the MongoDB Atlas dashboard for a cluster named 'localhost:27017' and a database named 'Sprint9'. At the top, there are buttons for 'Open MongoDB shell', 'Create collection', and 'Refresh'. Below these, a 'Sort by' dropdown is set to 'Collection Name'. The main area displays five collection cards:

- comments**: Storage size: 6.91 MB, Documents: 50 K, Avg. document size: 284.00 B, Indexes: 1, Total index size: 524.29 kB.
- movies**: Storage size: 22.23 MB, Documents: 24 K, Avg. document size: 1.60 kB, Indexes: 1, Total index size: 405.50 kB.
- sessions**: Storage size: 20.48 kB, Documents: 1, Avg. document size: 540.00 B, Indexes: 1, Total index size: 20.48 kB.
- theatres**: Storage size: 147.46 kB, Documents: 1.6 K, Avg. document size: 223.00 B, Indexes: 1, Total index size: 32.77 kB.
- users**: Storage size: 36.86 kB, Documents: 185, Avg. document size: 159.00 B, Indexes: 1, Total index size: 20.48 kB.

- Ejercicio 1

- Muestra los 2 primeros comentarios que aparecen en la base de datos.

The screenshot shows the MongoDB Atlas query editor for the 'comments' collection. The 'Documents' tab is selected, showing 50.3K documents. The query editor has a 'Type a query' field with a placeholder '{ field: 'value' } or Generate query'. Below this, the query is defined as:

```
Project: { field: 0 }
Sort: { date: 1 }
Collation: { locale: 'simple' }
Index Hint: { field: -1 }
```

On the right, there are buttons for 'Explain', 'Reset', 'Find', and 'Options'. Below the query editor, there are input fields for 'Max Time MS' (60000), 'Skip' (0), and 'Limit' (2). At the bottom, there are buttons for 'ADD DATA', 'EXPORT DATA', 'UPDATE', and 'DELETE'. The results section shows two documents:

```
{
  "_id": ObjectId("5a9427648b0beeb695e67e"),
  "name": "Mercedes Tyler",
  "email": "mercedes_tyler@fakegmail.com",
  "movie_id": ObjectId("573a1399f29313caabcec3ce"),
  "text": "Optio totam dolores magni. Enim ratione fuga tempora voluptatum est cu...",
  "date": 1970-01-01T01:07:09.000+00:00
}
```

```
{
  "_id": ObjectId("5a9427658b0beeb6969efe"),
  "name": "Jon Snow",
  "email": "kit_harington@gameofthron.es",
  "movie_id": ObjectId("573a13a4f29313caabd117df"),
  "text": "Dolorem animi tempora ullam quas. Iusto nobis reprehenderit aspernatur...",
  "date": 1970-01-01T09:37:49.000+00:00
}
```

Opcion2:

```
>_MONGOSH
> use Sprint9
< switched to db Sprint9
> db.comments.find().sort({date:1}).limit(2)
< {
  _id: ObjectId('5a9427648b0beeb695e67e'),
  name: 'Mercedes Tyler',
  email: 'mercedes_tyler@fakegmail.com',
  movie_id: ObjectId('573a1399f29313caabcec3ce'),
  text: 'Optio totam dolores magni. Enim ratione fuga tempora voluptatum est cumque animi. Quia nam doloribus corrupti id voluptate esse.
  date: 1970-01-01T01:07:09.000Z
}
{
  _id: ObjectId('5a9427658b0beeb6969efe'),
  name: 'Jon Snow',
  email: 'kit_harington@gameofthron.es',
  movie_id: ObjectId('573a13a4f29313caabd117df'),
  text: 'Dolorem animi tempora ullam quas. Iusto nobis reprehenderit aspernatur cupiditate a tempore. Commodi alias expedita dolore explic
  date: 1970-01-01T09:37:49.000Z
}
Sprint9>
```

- ¿Cuántos usuarios tenemos registrados?

Hay 185 usuarios registrados.

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the 'Sprint9' database. The main panel displays the 'users' collection, with a red circle highlighting the 'Documents' tab and the count '185'. The 'Query' tab is active, showing a filter of '{}'. Below the query, there are fields for 'Project', 'Sort', 'Collation', and 'Index Hint'. At the bottom, two document snippets are visible, each containing fields like '_id', 'name', 'email', and 'password'.

Opción2:

```
>_MONGOSH
> use Sprint9
< switched to db Sprint9
> db.users.countDocuments()
< 185
Sprint9>
```

- ¿Cuántos cines existen en el estado de California?

Hay 169 cines en California.

localhost:27017 > Sprint9 > theatres Open MongoDB shell

Documents 1.6K Aggregations Schema Indexes 1 Validation

Query: { "location.address.state": "CA" } Generate query Explain Reset Find </> Options

Project: { field: 0 }

Sort: { field: -1 } or [['field', -1]]

Collation: { locale: 'simple' }

Index Hint: { field: -1 }

Max Time MS: 60000

Skip: 0 **Limit:** 0

ADD DATA EXPORT DATA UPDATE DELETE 100 1 - 100 of 169 ↺ ↻ ⋮ {} ⌂

Results:

- `{ "_id": ObjectId("59a47286cfa9a3a73e51e72e"), "theaterId": 1008, "location": { ... } }`
- `{ "_id": ObjectId("59a47286cfa9a3a73e51e734"), "theaterId": 1009, "location": { ... } }`
- `{ "_id": ObjectId("59a47286cfa9a3a73e51e737"), "theaterId": 1018, "location": { ... } }`
- `{ "_id": ObjectId("59a47286cfa9a3a73e51e739"), "theaterId": 1011, "location": { ... } }`

- ¿Cuál fue el primer usuario en registrarse?

Ned Stark (Tiene el primer _id creado)

localhost:27017 > Sprint9 > users Open MongoDB shell

Documents 185 Aggregations Schema Indexes 1 Validation

Query: {} Generate query Explain Reset Find </> Options

Project: { field: 0 }

Sort: { _id: 1 }

Collation: { locale: 'simple' }

Index Hint: { field: -1 }

Max Time MS: 60000

Skip: 0 **Limit:** 1

ADD DATA EXPORT DATA UPDATE DELETE 100 1 - 1 of 1 ↺ ↻ ⋮ {} ⌂

Results:

- `{ "_id": ObjectId("59b99db4cfa9a34dcd7885b6"), "name": "Ned Stark", "email": "sean_bean@gameofthron.es", "password": "$2b$12$UREFwsRUoyF0CRqGNK0Lz00HM/jLhgUCNNIj9RJaqMUQ74crLj1Vu" }`

- ¿Cuántas películas de comedia existen en nuestra base de datos?

Existen 7024 películas de comedia.

The screenshot shows the MongoDB Compass interface. At the top, there's a navigation bar with tabs for 'movies', 'sessions', 'comments', 'users', 'coleccion1', 'theatres', 'Sprint9', and a plus icon. Below this, the breadcrumb 'localhost:27017 > Sprint9 > movies' is visible. The main panel has tabs for 'Documents' (23.5K), 'Aggregations', 'Schema', 'Indexes' (1), and 'Validation'. The 'Documents' tab is active, showing a query: `{ genres: "Comedy" }`. Below the query, there are fields for 'Project', 'Sort', 'Collation', and 'Index Hint'. To the right, there are buttons for 'Generate query', 'Explain', 'Reset', 'Find', and 'Options'. Further right, there are input fields for 'Max Time MS' (60000), 'Skip' (0), and 'Limit' (0). Below these fields, there are buttons for 'ADD DATA', 'EXPORT DATA', 'UPDATE', and 'DELETE'. At the bottom, there's a status bar showing '100' and '1 - 100 of 7024'. The main content area displays a JSON document for a movie: `{ "_id": ObjectId('573a1390f29313caabcd4803'), "plot": "Cartoon figures announce, via comic strip balloons, that they will mov...", "genres": Array (3), "runtime": 7, "cast": Array (1), "num_mflix_comments": 1, "poster": "https://m.media-amazon.com/images/M/MV5BYzg2NjNhNTctMjUxMi00ZWU4LWI3Zj...", "title": "Winsor McCay, the Famous Cartoonist of the N.Y. Herald and His Moving ...", "fullplot": "Cartoonist Winsor McCay agrees to create a large set of drawings that ...", "languages": Array (1), "released": 1911-04-08T00:00:00.000+00:00, "directors": Array (2) }`

- Ejercicio 2

Muéstrame todos los documentos de las películas producidas en 1932, pero que el género sea drama o estén en francés.

Hay un total de 18 documentos referentes a estos criterios de búsqueda.

The screenshot shows the MongoDB Compass interface with a query: `{year: 1932, $or:[{genres:"Drama"},{languages:"French"}]}`. The status bar at the bottom indicates '1 - 18 of 18'. The main content area displays two JSON documents. The first document is for a movie: `{ "_id": ObjectId('573a1392f29313caabcd9a76'), "plot": "In this romance a librarian takes a cruise and falls for an unobtainab...", "genres": Array (2), "runtime": 85, "rated": "APPROVED", "cast": Array (4), "num_mflix_comments": 3, "poster": "https://m.media-amazon.com/images/M/MV5BMjEzNTU5NTk2N15BM15BanBnXkFtZT...", "title": "Forbidden", "fullplot": "On a cruise to Cuba, Lulu Smith falls in love with Bob Grover. Back ho...", "languages": Array (2), "released": 1932-01-15T00:00:00.000+00:00, "directors": Array (1), "writers": Array (2), "awards": Object, "lastupdated": "2015-09-05 00:24:21.237000000", "year": 1932, "indb": Object, "countries": Array (1), "type": "movie", "tomatoes": Object }`. The second document is for a movie: `{ "_id": ObjectId('573a1392f29313caabcd9a7e'), "plot": "A circus' beautiful trapeze artist agrees to marry the leader of side-...", "genres": Array (2), "runtime": 64, "rated": "UNRATED", "cast": Array (4) }`

- Ejercicio 3

Muéstrame todos los documentos de películas estadounidenses que tengan entre 5 y 9 premios que fueron producidas entre 2012 y 2014.

localhost:27017 > Sprint9 > movies Open MongoDB shell

Documents 23.5K Aggregations Schema Indexes 1 Validation

```
{
  countries: "USA",
  year: { $lte: 2014, $gte: 2012 },
  "awards.wins": { $gte: 5, $lte: 9 }
}
```

Generate query Explain Reset Find Options

Project { title: 1, year: 1, countries: 1, awards: 1 }

Sort { year: 1, title: 1 }

Collation { locale: 'simple' }

Index Hint { field: -1 }

Max Time MS 60000

Skip 0 Limit 0

EXPORT DATA 100 1 - 100 of 166

```
_id: ObjectId('573a13c1f29313caabd65e49')
title: "21 Jump Street"
awards: Object
  year: 2012
  wins: 5
countries: Array (1)
```

```
_id: ObjectId('573a13d2f29313caabd91fb7')
year: 2012
title: "Ai Weiwei: Never Sorry"
```

Nivel 2

- Ejercicio 1

Cuenta cuántos comentarios escribe un usuario que utiliza "GAMEOFTHRON.ES" como dominio de correo electrónico.

Hay 22841 comentarios

{email:{\$regex: 'gameofthron\\.es\$'}}

Generate query Explain Reset Find Options

Project { field: 0 }

Sort {date:1}

Collation { locale: 'simple' }

Index Hint { field: -1 }

Max Time MS 60000

Skip 0 Limit 0

ADD DATA EXPORT DATA UPDATE DELETE 100 1 - 100 of 22841

```
_id: ObjectId('5a9427658b0beebe6969efe')
name: "Jon Snow"
email: "kit_harrington@gameofthron.es"
movie_id: ObjectId('573a13a4f29313caabd117df')
text: "Dolorem animi tempora ullam quas. Iusto nobis reprehenderit aspernatur..."
date: 1970-01-01T09:37:49.000+00:00
```

```
_id: ObjectId('5a9427658b0beebe697ad20')
name: "Sandor Clegane"
email: "rory_mccann@gameofthron.es"
movie_id: ObjectId('573a13dbf29313caabda14c')
text: "Voluptas quas in velit dolorum dolores. Veniam adipisci ullam mollitia..."
date: 1970-01-01T21:02:31.000+00:00
```

- Ejercicio 2

¿Cuántos cines existen en cada código postal situados dentro del estado Washington DC (DC)?

Existe 1 cine por cada uno de los 3 códigos postales de Washington presentes en la base de datos.

CONSULTA:

The screenshot shows the MongoDB Aggregations Builder interface for a collection named 'theatres'. The pipeline consists of three stages:

- Stage 1: \$match**
Query:

```
{ "location.address.state": "DC" }
```


Output: Sample of 3 documents showing theatre details for DC.
- Stage 2: \$group**
Query:

```
{ "_id": "$location.address.zipcode", "Total_cines": { $sum: 1 } }
```


Output: Sample of 3 documents showing the count of theatres per zipcode.
- Stage 3: \$sort**
Query:

```
{ "field1": 1 }
```


Output: Sample of 3 documents sorted by field1.

RESULTADO

The screenshot shows the final results of the aggregation pipeline. The results are displayed as a list of documents, each representing a zipcode and the count of theatres.

_id	Total_cines
"20002"	1
"20010"	1
"20016"	1

Nivel 3

- Ejercicio 1

Encuentra todas las películas dirigidas por John Landis con una puntuación IMDb (Internet Movie Database) de entre 7,5 y 8.

localhost:27017 > Sprint9 > movies Open MongoDB shell

Documents 23.5K Aggregations Schema Indexes 1 Validation

Query: `{directors: "John Landis", "imdb.rating": {$gte: 7.5, $lte: 8}}` Generate query Explain Reset Find Options

Project: `{title:1, directors:1, "imdb.rating":1}`

Sort: `{ field: -1 } or [['field', -1]]`

Collation: `{ locale: 'simple' }`

Index Hint: `{ field: -1 }`

Max Time MS: 60000

Skip: 0 Limit: 0

EXPORT DATA 100 1 - 4 of 4 100 1 - 4 of 4 100 1 - 4 of 4 100 1 - 4 of 4

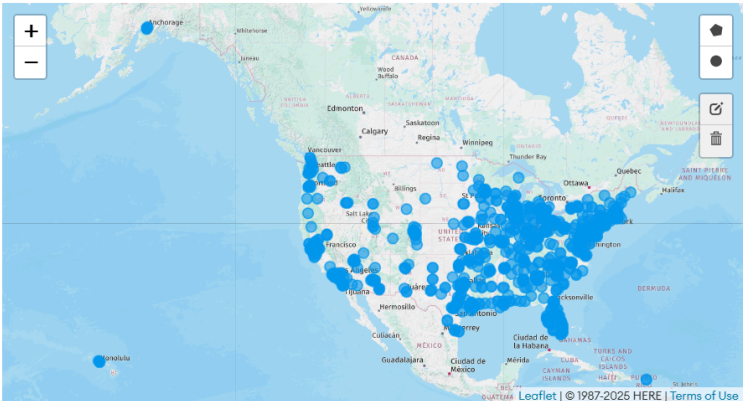
- `_id: ObjectId('573a1397f29313caabce6d94')`
 - `imdb: Object`
 - `title: "Animal House"`
 - `directors: Array (1)`
- `_id: ObjectId('573a1397f29313caabce76f7')`
 - `title: "The Blues Brothers"`
 - `directors: Array (1)`
 - `imdb: Object`
- `_id: ObjectId('573a1397f29313caabce7d06')`
 - `imdb: Object`
 - `title: "An American Werewolf in London"`
 - `directors: Array (1)`
- `_id: ObjectId('573a1398f29313caabce8deb')`
 - `title: "Trading Places"`

- Ejercicio 2

Muestra en un mapa la ubicación de todos los teatros de la base de datos.

coordinates

coordinates



The map displays the United States and parts of Canada and Mexico. Numerous blue dots are scattered across the landmasses, representing the locations of theaters. The dots are most densely clustered in the eastern half of the United States, particularly in the Northeast and Midwest, and are more sparsely distributed in the western half. The map includes a scale bar, a compass rose, and a legend.